

Strict partition chains and their asymptotic expansions

Report 223

Abstract

The Lengyel numbers, OEIS A005121, count strict chains between the two endpoints of a partition lattice. We give a self-contained recurrence proof of their complete fixed-order asymptotic expansion on the known factorial-square scale. An elementary factorial majorant and a finite block-defect expansion yield a uniform kernel remainder. A weighted first-difference operator has limiting norm $2\log 2 - 1 < 1$, so recurrence residuals give genuine asymptotic errors and one positive, anchored leading constant. The same argument on a fixed complex marking disk proves all fixed cumulant expansions for chain length. We also give a constructive Lambert- W inverse, eventual recovery on sampled values, and two-ceiling enclosures for arbitrary thresholds. Exact counting and testing code accompanies the article. The leading equivalent, its constant, the leading mean, and the central limit theorem are prior results. Formal higher corrections were previously announced; no priority claim for these coefficients or general inversion machinery is made.

[write] **Status and trust boundary, added 5 October 2026, batch 106.** This is bundle Report 223 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a005121-strict-partition-chains** (provenance, the sources as the write read them, relation to the repository, reading conventions and the collected non-claims in Section 1.2). It is unrefereed and not formalized; its author line reads “Report 223” and its PDF author field is empty. *What rests on what.* Theorems 1.1 and 8.1 and everything derived from them are proved by hand in Sections 3–10 from the recurrence (1); no proof uses a computation or a numerical value of C . The decimals of Section 11 ($C \approx 1.098685805525187$, K_1 , K_2) are uncertified diagnostics. The leading equivalent, the positivity of C , the leading mean and the central limit theorem are re-proofs of results of Babai–Lengyel and Van Cutsem–Ycart, as the source says. The write adds two remarks with proofs: Remark 2.1 (the generating-function formula printed in OEIS A005121 has the wrong sign; its only formal solution is the generating function of $-z_n$) and Remark 10.5 (which parts of Section 10 are instances of the transseries volume’s theorems, and a second route to Theorem 10.4); and Section 12.1, the open questions, with a numerical observation on the depth-sum route through **a139383-iterated-bell-diagonals**.

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1 Objects and the main conclusions

Let Π_n be the lattice of set partitions of $[n] = \{1, \dots, n\}$, ordered by refinement. Its least element consists of n singleton blocks, and its greatest element has one block. A *strict chain* here starts at the least element, ends at the greatest element, and has strictly increasing successive partitions. It need not be saturated. Its length H is the number of strict transitions, not the number of partitions listed. In particular $H_1 = 0$; for $n \geq 2$, $1 \leq H \leq n - 1$.

Let z_n be the number of these chains, and let $Z_n(q)$ count them with weight q^H . If $S(n, j)$ denotes a Stirling number of the second kind, then

$$Z_1(q) = 1, \quad Z_n(q) = q \sum_{j=1}^{n-1} S(n, j) Z_j(q) \quad (n \geq 2), \quad z_n = Z_n(1). \quad (1)$$

Indeed, the first transition has a partition with $j < n$ blocks, of which there are $S(n, j)$ possibilities. Its upper interval is a partition lattice on those j blocks. This proves the recurrence without any independence assumption. The first values are

$$(z_1, \dots, z_{10}) = (1, 1, 4, 32, 436, 9012, 262760, 10270696, 518277560, 32795928016).$$

At $q = 1$ choose a chain uniformly. Its probability generating function is exactly $Z_n(q)/z_n$.

Write throughout

$$L = \log 2, \quad b = 1 + \frac{L}{3}, \quad f_n = (n!)^2 (2L)^{-n} n^{-b}. \quad (2)$$

Theorem 1.1 (Unmarked fixed-order expansion). *There is a positive constant C , uniquely anchored by*

$$C = \lim_{n \rightarrow \infty} \frac{z_n}{f_n}, \quad (3)$$

such that for every fixed integer $M \geq 0$,

$$z_n = C f_n \left(\sum_{j=0}^M \frac{c_j(L)}{n^j} + O_M(n^{-M-1}) \right), \quad c_0 = 1. \quad (4)$$

The coefficients $c_j \in \mathbb{Q}[L]$ are uniquely and recursively computable. The first four are

$$\begin{aligned} c_1 &= \frac{L}{6}, \\ c_2 &= -\frac{L(16L^2 + 45L - 90)}{3240}, \\ c_3 &= \frac{L^2(10L^2 - 81L - 270)}{6480}, \\ c_4 &= \frac{L}{146966400} (1792L^5 + 32400L^4 + 2358405L^3 \\ &\quad - 1281420L^2 - 8108100L - 408240). \end{aligned} \quad (5)$$

The proof in Sections 3–7 is direct and does not invoke a previously known convergence theorem. This does not make the leading limit new: it is the known Lengyel constant. The point is that the error transfer is proved, rather than inferred from formal substitution.

There is also one complex neighborhood of $q = 1$ on which the same polynomials give every fixed-order marked expansion. The initial contraction disk can be taken to be $|q - 1| \leq 1/16$; a once-only further shrink makes the analytic limiting constant nonzero. With $L(q) = \text{Log}(q + 1) - \text{Log } q$,

$$Z_n(q) = C(q)(n!)^2 [2L(q)]^{-n} n^{-1-L(q)/3} \left(\sum_{j=0}^M \frac{c_j(L(q))}{n^j} + O_M(n^{-M-1}) \right). \quad (6)$$

The error inside parentheses is analytic and uniform on a fixed disk. Its fixed derivatives, on smaller disks, retain the stated order. This is a statement about a normalized remainder; differentiating the whole prefactor would introduce additional n -dependent factors.

In particular, for constants K_1, K_2 determined by derivatives of $\log C(e^t)$,

$$\mathbb{E}H_n = \frac{n}{2L} + \frac{\log n}{6} + K_1 - \frac{1}{12n} + O(n^{-2}), \quad (7)$$

$$\text{Var } H_n = \frac{1-L}{4L^2} n - \frac{\log n}{12} + K_2 + \frac{1}{24n} + O(n^{-2}). \quad (8)$$

Every fixed cumulant has a full fixed-order expansion of the same kind. Section 10 gives the corresponding inverse construction and explains precisely why arbitrary threshold inversion needs an enclosure rather than the ceiling of an unqualified approximation.

1.1 What the asymptotic assertions mean

All orders in this article are *fixed*. For each chosen M there are constants and an onset index for the indicated remainder. Neither their sizes nor the onset need be uniform as M grows. We prove no convergence of the infinite formal series, Gevrey estimate, optimal truncation, exponentially small remainder, local limit theorem, Edgeworth expansion, or large-deviation principle. No interval-certified decimal value of C or its marked derivatives is supplied. Exact calculations and decimal diagnostics are separated in Section 11.

1.2 Provenance, sources and reading conventions

[write] **Added 5 October 2026, batch 106.** *Provenance.* This report is Report 223 of the session bundle of Reports 1–243 that ProveIt’s docs/incoming intake processed as batch 106: the archive **Report223.zip** (509,137 bytes, SHA-256 `c178191805f9...cf2b8f25`, 15 files at the archive root, no wrapper directory), arrival commit `60f54ea06`, placed in `47fc7a069`. The text is the delivered **Report223.tex** (787 lines), shipped as **article.tex**. The delivered 19-page PDF and the checksum manifest **SHA256SUMS** are not shipped (the manifest was verified, 14 of 14 entries, at placement); the delivered README is replaced by the report README; all three are retrievable from the arrival commit. The manuscript is undated. Its pin is ProveIt commit `9f19e58de`: it inspected the article of **a139383-iterated-bell-diagonals** there (blob `ab5014ec`, which is still the blob at the commit of this write, so what the source says about that report concerns its current text) and the transseries catalogue. It carries no “prepared for private review” line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 78 delivered labels with **spc**: and updated the 59 references to them; it added this subsection, the notes marked [write], Remarks 2.1 and 10.5, Section 12.1 and a note in one bibliography entry. No statement, proof or number of the source was changed: the added remarks are the last statements of their sections and the added displays are unnumbered, so every theorem and equation keeps its delivered number.

The sources, as the write read them (5 October 2026).

- OEIS A005121 [8], live entry: revision #79 of May 30, 2026, the revision the source inspected, with 19 terms through $z_{19} = 879518067472225603327860638128$. Its formula line “E.g.f. satisfies $Z(z) = 1/2 * (Z(\exp(z)-1) - z)$. [Lengyel]” is discussed in Remark 2.1. Its asymptotic line attributes the leading equivalent to Babai and Lengyel and the constant $C_L = 1.0986858055\dots$ (A086053) to Flajolet and Salvy.
- **a139383-iterated-bell-diagonals** of this repository: the article’s Part II, Section 17.2 (“Other overlap and scope restrictions”; lines 909–910 of its **article.tex**) states the depth-sum identity $Z_n = \sum_{m \geq 0} 2^{-m-1} H(n, m)$ from $2Z(z) = z + Z(e^z - 1)$ (the correct sign) and that “a depth-summed equivalent alone does not give a pointwise proportional-depth equivalent”; its delivered source note **28-proportional-sources-literature.md** (lines 139–141) records the known Lengyel asymptotic. Read in full where cited.
- The transseries volume **Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex**: `p0:def:core`, `p0:prop:factorial-core`, `p0:thm:lambert-core`, `p0:thm:core-reversion`, `p0:thm:backward-error`, `p0:def:three-inverses` and `p0:thm:staircase`, read for Remark 10.5.
- Not read by the write: Lengyel [1], Babai and Lengyel [2], Prellberg [3], Mishna [4], Van Cutsem and Ycart [5], Finch [6] and Dickey and Rosenberg [7]. The statements made about them in Section 2 are the source’s; **data/SOURCES.json** records what the source inspected, with SHA-256 fingerprints of the files it read. In particular, whether Lengyel’s 1984 paper prints the generating-function equation with the sign of the OEIS line was not checked.

What was checked at intake. The batch-106 dossier read the manuscript in full and found no error in it. Independently of the delivered code it computed z_n by the recurrence (1) to $n = 500$: $z_1, \dots, z_{10}$ agree with the OEIS, and $z_n / (f_n(1 + c_1/n + c_2/n^2 + c_3/n^3))$ equals 1.0986858052222, 1.0986858055062, 1.0986858055214, 1.0986858055240, 1.0986858055247 at $n = 100, 200, 300, 400, 500$, within about $5 \cdot 10^{-13}$ of the source’s 1.098685805525187 at $n = 500$, which confirms c_1, c_2, c_3 and the decimal of C (an observation, not a certificate). It checked the induction of Lemma 3.1 and confirmed the OEIS sign discrepancy in the live entry. The write rechecked identity (11) from the Stirling recurrence, the coefficient sums $Z_n(1) = z_n$ of the six listed chain polynomials, and the OEIS history of the formula line (Remark 2.1). The delivered programs passed on copies at placement and at the write (Section 11, note at the end).

Relation to the repository. Before batch 106 the repository mentioned A005121 only in **a139383-iterated-bell-diagonals**, as background (the two places above); no file mentioned A086053 or A008826. That report is not a host for this one: it answers no question of this report, this report answers none of its questions, and the methods differ (here a Stirling-kernel recurrence with a first-difference contraction; there parabolic iteration of $e^z - 1$ and a fixed Fatou-coordinate contour). The source’s own Question 6 (Section 12) asks for

the comparison; it is Question 6 below, with a numerical observation. The source’s bibliography credits that report to “V. Reshetnikov”; the report has no author line (its title block reads “A merged research report built from two manuscripts”), so it is cited here by repository path (note at [9]). The inverse construction of Section 10 and the transseries volume are compared in Remark 10.5: its Lambert core is an instance of the volume’s `p0:prop:factorial-core` (not of `p0:thm:lambert-core`). No Lean or Rocq file of the repository treats partition-lattice chains; no statement of this report is formalized, and its place in the collection confers no formal status.

[write] What is not claimed, collected from the source. All orders are fixed: no convergence of the formal series, no Gevrey estimate, optimal truncation or exponentially small remainder, no local limit theorem, Edgeworth expansion or large-deviation principle, no distributional error estimate. No interval-certified value of C , K_1 , K_2 or of the marked derivatives; no numerically efficient constant A_M in (32); the constants A_M , Y_M of Theorem 10.4 are not evaluated, and the exact inverse in the code is bounded by its declared resources. The leading equivalent, the constant, the leading mean and the central limit theorem are prior results; no priority is claimed for the correction coefficients (formal higher corrections were announced by Prellberg; the 1990 Flajolet–Salvy manuscript was not recovered), for the inversion machinery, or for the resolution of Van Cutsem and Ycart’s general variance conjecture, which is not resolved. The disk $|q - 1| \leq 1/16$ is convenient, not optimal, and gives no global marked statement. The generating function is formal (radius zero). The source’s comparison with the literature is “an inspected source boundary, not a worldwide-priority certificate”; that the ProveIt iterated-Bell results cannot imply aggregate statements is not claimed. Cross-version byte identity of the rebuild is not promised. The write adds: its numerical observation in Question 6 is a floating-point extrapolation and certifies nothing.

Reading conventions (letters the manuscript reuses, with the tempting false readings; no symbol was renamed).

Symbol	Meaning here (where)	Not to be read as
$z_n, Z_n(q)$	strict endpoint chains of Π_n (A005121), and their length polynomial; $z_n = Z_n(1)$	a139383's Z_n is the same number, written as a depth sum
$\mathcal{Z}(z)$	the formal e.g.f. $\sum z_n z^n / n!$ (Section 2)	the OEIS's $Z(z)$, which obeys the misprinted equation (Remark 2.1)
H, H_n	chain length (number of strict transitions; $H_1 = 0$)	a139383's iterated Bell numbers $H(n, m)$; $H(x)$ and $H_{e,k}(t)$ below
$H(x), H_{e,k}(t)$	$2x \log(x/e^\kappa)$ (Section 10); a Taylor function (Lemma 4.1)	the length H_n
$L, L(q)$	$\log 2$; $\text{Log}(q+1) - \text{Log } q$, so $L(1) = L$	a target value (the transseries volume writes L for the target)
b	$1 + L/3$	the coefficient b of $aX + b \log X$ in <code>p0:thm:lambert-core</code>
$C, C(q), C_M, C_e(k)$	Lengyel's constant; its marked analytic version; the order- M limits (all equal to C); block-configuration counts (Section 4)	each other
$K(n, k); K; K_1, K_2$	the normalized kernel; $2\pi C$ (Section 10); the mean and variance constants	each other
$T_r(L), T_r(n)$	Touchard polynomials (26); the weighted norm (29) in Lemma 6.1	each other
$q, q_j(k; L)$	the marking variable; the kernel polynomials (18)	
d, d_n, d_j	a block count (Section 4); first differences (Lemma 6.1); logarithmic coefficients (47)	each other
e	Euler's number; the block excess in Section 4	
κ, κ_r	$1 + \frac{1}{2} \log(2L)$ (Section 10); the r th cumulant	the slope $\kappa = 2$ of <code>p0:prop:factorial-core</code> (Remark 10.5)
β, μ	$L/3$; the kernel mean $2L$	a139383's slope parameter $\beta = n/m$
$A, B, R; A(t), B(t), D(t)$	constants of Lemma 6.1; cumulant functions (43)	each other; the disks D_0, D
Λ, w, X, U, v_m	$2(1+w)$; $W_0(y/(2e^\kappa))$; $e^{\kappa+w}$; $\log X$; inverse coefficients (50)	Λ is the core slope of the transseries volume (same meaning)
$N(y)$	$\min\{n \geq 2 : \log z_n \geq y\}$	a threshold on z_n itself (the input is log-scale)

2 Prior results and the scope of this report

Lengyel [1] proved two-sided bounds on the scale (2); his Theorem 1.2 announced a convergence result. Babai and Lengyel [2], Theorem 2, proved the positive limit (3) using a general nearly convex recurrence criterion. The end of their Section 2 explicitly asks about convergence speed. Their work also supplies the proper historical setting for the recurrence method. Near-diagonal Stirling expansions, including the fixed-deficit polynomial structure, have earlier origins; Lengyel's discussion credits Hsu.

Prellberg’s 2002 presentation [3] gives the leading equivalent and a contour expression for the constant by analytic iteration. Its final slide describes additional coefficient computations as using a separate, non-rigorous method. Mishna’s seminar summary [4] records the leading chain asymptotic and the analytic-iteration approach. Thus a rigorous fixed-order remainder proof must be distinguished from an assertion that higher coefficients were previously unknown.

Finch [6], Section 5.7, credits an unfinished 1990 Flajolet–Salvy manuscript with computation of the constant and analytic-iteration work. That full manuscript was not available for this comparison. This unresolved source prevents a claim about the first occurrence of the correction coefficients or an exhaustive priority determination.

Van Cutsem and Ycart [5], Theorem 4.1 and Example 1, give the leading mean behavior and a central limit theorem for the strict partition-chain model. Their printed page 997 distinguishes those conclusions from a variance equivalent conjectured under their broader hypotheses. The present marked proof concerns only recurrence (1). It is not a resolution of that general conjecture. Our central limit argument is a reproof of a known result; the complex-uniform fixed-order expansion is what supports the detailed cumulant corrections here.

Related proportional-depth iterated-Bell results already appear in the ProveIt article [9]. The actual article, at the pinned revision stated in the references, was compared. It discusses A005121 as a weighted depth aggregate and proves all-order results for a different, related observable. Those results are relevant context; no claim is made that they cannot imply aggregate statements. The route here is a direct recurrence argument for the aggregate and its chain-length marking. Likewise the inverse method specializes the established Lambert- W and residual-transport apparatus already represented in ProveIt’s transseries work [10]; it is not a new general inverse theory.

A recent phylogenetic interpretation is given by Dickey and Rosenberg [7]: the simultaneous-merger diagonal in their model is A005121. The arXiv article was compared for its enumerative scope; the published record is included bibliographically. These remarks describe an inspected source boundary, not a worldwide-priority certificate.

2.1 A formal generating function and a checked source discrepancy

Set $\mathcal{Z}(z) = \sum_{n \geq 1} z_n z^n / n!$ as a *formal* series. Adding the diagonal term $S(n, n)z_n = z_n$ to (1) gives

$$2\mathcal{Z}(z) = z + \mathcal{Z}(e^z - 1). \quad (9)$$

The coefficient of z on both sides is 2. The OEIS raw source record [8], revision #79 of 30 May 2026, instead displays a minus sign before z in the equivalent half-formula. That sign conflicts already with $z_1 = 1$; equation (9) follows immediately from the recurrence. This report records the discrepancy without editing the entry.

Equation (9) is not an analytic equation for a convergent exponential generating function at the origin. The proved growth gives

$$\left(\frac{z_n}{n!}\right)^{1/n} \sim \frac{n}{2eL} \longrightarrow \infty,$$

so its radius of convergence is zero. Formal composition is nevertheless well defined because $e^z - 1$ has zero constant term.

Remark 2.1 ([write] The sign in OEIS A005121, 5 October 2026). The live OEIS entry A005121 [8] (revision #79 of May 30, 2026, read 5 October 2026) has the formula line

$$\text{E.g.f. satisfies } Z(z) = 1/2 * (Z(\exp(z)-1) - z). \quad [\text{Lengyel}]$$

The line carries no contributor's signature or date. In the entry's edit history it first appears in revision #8 of September 13, 2003 (an edit by N. J. A. Sloane, together with the asymptotic line), attributed "(Lengyel)"; revision #30 of April 18, 2014 changed the attribution to "[Lengyel]"; the sign has not changed since 2003. With $\mathcal{Z}(z) = \sum_{n \geq 1} z_n z^n / n!$ the correct equation is (9), that is $\mathcal{Z}(z) = \frac{1}{2}(\mathcal{Z}(e^z - 1) + z)$: the OEIS line has $-z$ where $+z$ is right.

Proof. For any formal series $W(z) = \sum_{n \geq 0} w_n z^n / n!$, the expansion $(e^z - 1)^k / k! = \sum_{n \geq k} S(n, k) z^n / n!$ gives $W(e^z - 1) = w_0 + \sum_{n \geq 1} (\sum_{k=1}^n S(n, k) w_k) z^n / n!$. Hence the equation $2W(z) = W(e^z - 1) + \varepsilon z$, with $\varepsilon = \pm 1$, holds if and only if $2w_0 = w_0$ and

$$2w_n = \sum_{k=1}^n S(n, k) w_k + \varepsilon [n = 1] \quad (n \geq 1),$$

that is

$$w_0 = 0, \quad w_1 = \varepsilon, \quad w_n = \sum_{k=1}^{n-1} S(n, k) w_k \quad (n \geq 2),$$

using $S(n, n) = 1$. So each sign has exactly one formal solution, and by (1) at $q = 1$ it is $w_n = \varepsilon z_n$. For $\varepsilon = +1$ this is $\mathcal{Z}$, proving (9). For $\varepsilon = -1$, the OEIS line, the only solution is $-\mathcal{Z}$, the generating function of $(-1, -1, -4, -32, -436, \dots)$. For $\mathcal{Z}$ itself the right-hand side of the printed line is $\frac{1}{2}(\mathcal{Z}(e^z - 1) - z) = \mathcal{Z}(z) - z$ by (9), so the line fails exactly at the coefficient of z , where it would require $z_1 = \frac{1}{2}(z_1 - 1)$, that is $1 = 0$. Equivalently, $-\mathcal{Z}$ satisfies the printed line, and $\mathcal{Z}$ does not. (Wording sharpened after the independent check of 5 October 2026; the first wording read "already the coefficient of z in the OEIS line reads $1 = \frac{1}{2}(1 - 1) = 0$ for $\mathcal{Z}$ ".) $\square$

The same entry's PARI program (Michael Somos, September 22, 2007) iterates, with series truncations, `A = x - A + subst(A, x, exp(x)-1)`, whose fixed point satisfies $2A = x + A(e^x - 1)$, the correct sign, and `a139383-iterated-bell-diagonals` (Part II, Section 17.2) prints $2Z(z) = z + Z(f(z))$ with $f(z) = e^z - 1$, also correct. The source records the discrepancy at the raw OEIS record of the same revision (above); it is recorded here only, and nothing was submitted to the OEIS.

3 An elementary global kernel bound

Put $(n)_k = n(n-1) \cdots (n-k+1)$ and define

$$a(n, k) = \frac{2^k S(n, n-k)}{(n)_k^2} \quad (0 \leq k \leq n-1), \quad a(n, 0) = 1,$$

with value zero outside the natural range when used in recurrences. The expression $(n)_k^2$ means $((n)_k)^2$.

Lemma 3.1 (Factorial domination). *For $0 \leq k \leq n-1$,*

$$0 \leq a(n, k) \leq \frac{1}{k!}. \quad (10)$$

Proof. The Stirling recurrence gives the exact identity

$$a(n, k) = \frac{2(n-k)}{n^2} a(n-1, k-1) + \frac{(n-k)^2}{n^2} a(n-1, k). \quad (11)$$

The boundary term outside the natural range is zero. Starting with $a(1, 0) = 1$, induction and nonnegativity imply

$$k!a(n, k) \leq \frac{2k(n-k) + (n-k)^2}{n^2} = 1 - \frac{k^2}{n^2} \leq 1.$$

The case $k = 0$ is immediate. $\square$

The normalized kernel and its limiting weights are

$$K(n, k) = S(n, n-k) \frac{f_{n-k}}{f_n} = L^k a(n, k) (1 - k/n)^{-b}, \quad p_k = \frac{L^k}{k!} \quad (k \geq 1). \quad (12)$$

Lemma 3.1 yields

$$K(n, k) \leq 2^b p_k \quad (k \leq n/2), \quad K(n, k) \leq n^b p_k \quad (1 \leq k < n). \quad (13)$$

For every fixed $\varepsilon > 0$, real A , and integer $d \geq 0$,

$$n^A \sum_{\varepsilon\sqrt{n} < k < n} (1+k)^d K(n, k) \longrightarrow 0. \quad (14)$$

To check this explicitly, bound $(1+k)^d$ and n^b by powers of n . For large m the successive ratios in $L^k/k!$ for $k \geq m$ are at most $1/2$, so its tail is at most $2L^m/m!$. With $m \asymp \sqrt{n}$ and $m! \geq (m/e)^m$, this is $\exp(-c\sqrt{n} \log n)$ for some $c > 0$, which beats every fixed inverse power. The same tail argument applies to any fixed polynomial in k times p_k .

4 A uniform expansion by block defects

A fixed- k Stirling expansion alone would not justify summing a recurrence with k up to $n-1$. The following construction supplies both the coefficients and the uniform remainder.

4.1 An exact finite decomposition

For a partition with deficit k , let m_j be the number of its blocks of size $j \geq 2$. Define

$$e = \sum_{j \geq 3} (j-2)m_j, \quad d = \sum_{j \geq 3} (j-1)m_j, \quad m_2 = k - d.$$

The number of vertices in nonsingleton blocks is $2k - e$. Counting labeled blocks gives exactly

$$k!a(n, k) = \sum_{e \geq 0} C_e(k) \frac{\binom{n}{2k-e}}{((n)_k)^2}, \quad C_e(k) = \sum_{\substack{m_j \geq 0 \\ \sum_{j \geq 3} (j-2)m_j = e}} \frac{2^d (k)_d}{\prod_{j \geq 3} m_j! (j!)^{m_j}}. \quad (15)$$

Terms with $d > k$ vanish at integer k , so only legitimate block configurations contribute. For example,

$$C_0 = 1, \quad C_1 = \frac{2}{3}(k)_2, \quad C_2 = \frac{1}{3}(k)_3 + \frac{2}{9}(k)_4.$$

For a nonzero term with $e \geq 1$, $e+1 \leq d \leq \min(2e, k)$. There are at most 2^e configurations (integer partitions of e are no more numerous than compositions), and therefore

$$0 \leq C_e(k) \leq (8k^2)^e. \quad (16)$$

Moreover, for $k \leq n/2$ and a legitimate numerator,

$$\frac{\binom{n}{2k-e}}{\binom{n}{k}^2} \leq n^{-e} \exp(2k^2/n). \quad (17)$$

Indeed, use n^{2k-e} for the numerator and $-2 \sum_{i < k} \log(1 - i/n) \leq 4 \sum_{i < k} i/n \leq 2k^2/n$ for the denominator.

4.2 The complex Taylor estimate

Lemma 4.1 (Uniform kernel expansion). *For every fixed integer $J \geq 0$ there are polynomials $q_j(k; L) \in \mathbb{Q}[L, k]$, of degree at most $2j$ in k , such that uniformly for $1 \leq k \leq \sqrt{n}/8$,*

$$K(n, k) = p_k \left(\sum_{j=0}^J q_j(k; L) n^{-j} + O_J((1+k)^{2J+2} n^{-J-1}) \right). \quad (18)$$

Here $q_0 = 1$ and $q_j(0; L) = 0$ for $j \geq 1$. The same result is uniform when L ranges over a fixed compact complex neighborhood of $\log 2$ and $b = 1 + L/3$.

Proof. For every relevant e , set

$$H_{e,k}(t) = \frac{\prod_{i=0}^{2k-e-1} (1 - it)}{\prod_{i=0}^{k-1} (1 - it)^2} (1 - kt)^{-b}, \quad R_k = \frac{1}{8k^2}.$$

The branches at zero are used. On $|t| \leq R_k$ all factors are nonzero, and their arguments it have modulus at most $1/4$. Since $|\log(1 - z)| \leq (4/3)|z|$ for $|z| \leq 1/4$, summing the logarithms gives

$$|\log H_{e,k}(t)| \leq \frac{3 + |b|}{6}, \quad |H_{e,k}(t)| \leq B := \exp((3 + |b|)/6).$$

For clarity, the sum of numerator indices is at most $2k^2$, twice the denominator sum is at most k^2 , and the marking term contributes at most $|b|k$. This proves the stated bound uniformly in the relevant e and k .

Write $H_{e,k}(t) = \sum_{r \geq 0} A_{e,r}(k) t^r$. Since $8k^2/n \leq 1/8$, Cauchy's estimate yields, for every fixed $D \geq 0$,

$$\left| H_{e,k}(1/n) - \sum_{r=0}^D A_{e,r}(k) n^{-r} \right| \leq \frac{8B}{7} 8^{D+1} k^{2D+2} n^{-D-1}. \quad (19)$$

For $e \leq J$ take $D = J - e$ and multiply by $C_e(k) n^{-e}$. Equation (16) makes every resulting remainder $O_J(k^{2J+2} n^{-J-1})$. The remaining defect tail is bounded by

$$\sum_{e \geq J+1} C_e(k) \frac{\binom{n}{2k-e}}{\binom{n}{k}^2} \leq e^{2/64} \frac{(8k^2/n)^{J+1}}{1 - 8k^2/n}.$$

The extra factor $(1 - k/n)^{-b}$ is uniformly bounded, completing the remainder proof.

Finally, for $r \geq 1$ the coefficient of t^r in $\log H_{e,k}$ is

$$g_{e,r}(k) = \frac{2 \sum_{i=0}^{k-1} i^r - \sum_{i=0}^{2k-e-1} i^r + b k^r}{r}. \quad (20)$$

For fixed e this is a polynomial in k of degree at most $r + 1$. Exponentiation gives $\deg_k A_{e,r} \leq 2r$, while $\deg C_e \leq 2e$. Thus

$$q_j(k; L) = \sum_{e=0}^j C_e(k) A_{e,j-e}(k) \quad (21)$$

has degree at most $2j$. Polynomial power sums may be used when defining these polynomials: terms that would have a negative falling-factorial length at a small integer k have $C_e(k) = 0$ and contribute nothing. The finitely many such small k can equivalently be handled directly. For $e > 0$, C_e has a factor k ; the $e = 0, k = 0$ series is 1. Hence $q_j(0; L) = 0$ for $j > 0$. All bounds use only a bound for $|b|$, proving the complex-uniform version as well. $\square$

The first coefficients are

$$\begin{aligned} q_1(k; L) &= \frac{k(L - k + 1)}{3}, \\ q_2(k; L) &= \frac{k^2}{18} (L^2 - 2Lk + 5L + k^2 - 6k + 5). \end{aligned} \quad (22)$$

Equations (13) and (14) allow the expansion to be summed term by term through any fixed order with any fixed polynomial weight in k . In particular,

$$\sum_{k \geq 1} p_k = 1, \quad \mu := \sum_{k \geq 1} k p_k = 2L, \quad \sum_{k \geq 1} p_k q_1(k; L) = 0, \quad \sum_{k \geq 1} p_k q_2(k; L) = -\frac{L^2}{3}. \quad (23)$$

5 A triangular construction of every coefficient

Set $P_M(n) = \sum_{s=0}^M c_s n^{-s}$ with $c_0 = 1$. We choose the coefficients so that

$$\sum_{k=1}^{n-1} K(n, k) P_M(n - k) - P_M(n) = O_M(n^{-M-2}). \quad (24)$$

To see why this is constructive, for $s \geq 1$ expand

$$(n - k)^{-s} = n^{-s} \sum_{h \geq 0} \binom{s + h - 1}{h} (k/n)^h$$

only through the required finite order; for $s = 0$ just the $h = 0$ term is present. On the near range this finite Taylor expansion has a uniform remainder; on the far range (13) controls the bounded finite sequence values after the patch described below. Lemma 4.1 through order $M + 1$ then gives a residual expansion through order n^{-M-1} with error $O_M(n^{-M-2})$.

The constant term vanishes because $\sum p_k = 1$. The coefficient of n^{-1} vanishes because the q_1 mean is zero. At order n^{-m-1} the first uncanceled occurrence of c_m has multiplier

$$\sum_{k \geq 1} p_k (q_1(k; L) + mk) = m\mu = 2mL \neq 0. \quad (25)$$

Thus c_m is minus the contribution of the already known coefficients, divided by $2mL$. The contribution of c_{m+1} in a longer formal series would cancel between the leading kernel and the subtracted $P_M(n)$. There is consequently no hidden higher unknown in this step. At the first nontrivial order the residual is $-L^2/3 + 2Lc_1$, proving $c_1 = L/6$.

For an explicit finite algebraic recipe, define the moment functional on polynomials in k by

$$\mathcal{M}(1) = 1, \quad \mathcal{M}(k^r) = 2T_r(L) \quad (r \geq 1), \quad T_r(L) = \sum_{h=0}^r S(r, h)L^h. \quad (26)$$

Form $q_0, \dots, q_{m+1}$ using (20)–(21); finite exponential coefficients can be generated by

$$A_{e,0} = 1, \quad A_{e,r} = \frac{1}{r} \sum_{a=1}^r a g_{e,a} A_{e,r-a}.$$

Then take the coefficient of t^{m+1} in

$$\mathcal{M} \left[\left(\sum_{a=0}^{m+1} q_a(k; L) t^a \right) \left(\sum_{s=0}^m c_s t^s (1 - kt)^{-s} \right) \right] - \sum_{s=0}^m c_s t^s, \quad (27)$$

and set it to zero. Every operation is finite and rational.

Proposition 5.1. *The construction gives $c_m \in \mathbb{Q}[L]$ at every order, not merely $\mathbb{Q}(L)$.*

Proof. Each nontrivial polynomial contribution to a positive-order residual vanishes at $k = 0$: it contains either q_a for $a > 0$ or a positive shift power k^h . Every $T_r(L)$ for $r > 0$ is divisible by L . Hence the known residual contribution is divisible by L whenever earlier c_s lie in $\mathbb{Q}[L]$. Division by $2mL$ in (25) stays in $\mathbb{Q}[L]$. Induct from $c_0 = 1$. $\square$

This produces (5) and every further fixed coefficient. In subsequent recurrence arguments, redefine $P_M(n) = 1$ for the finitely many small integer indices where necessary. Since $P_M(n) \rightarrow 1$, this makes it positive and bounded above and below. A change at fixed $n - k$ has $k \geq n - O(1)$, so (13) makes its effect smaller than every inverse power. The residual (24) is unchanged asymptotically.

6 Strict contraction of first differences

The formal coefficient calculation is not yet an asymptotic theorem. The following transfer lemma supplies the needed stability and limiting constant.

Lemma 6.1 (Positive first-difference transfer). *Suppose $u_n > 0$ satisfies*

$$u_n = \sum_{k=1}^{n-1} w(n, k) u_{n-k}, \quad w(n, k) \geq 0, \quad \sigma_n := \sum_k w(n, k) = 1 + O(n^{-r}), \quad r > 1.$$

Suppose, for fixed constants $A, B, R > 0$,

$$w(n, k) \leq A \frac{R^k}{k!} \quad (k \leq n/2), \quad w(n, k) \leq A n^B \frac{R^k}{k!} \quad (k < n),$$

and $w(n, k) \rightarrow p_k$ at every fixed k , where $\sum p_k = 1$ and $\mu = \sum k p_k < 2$. Then u_n is bounded and bounded away from zero, converges to some $C > 0$, and

$$u_n = C + O(n^{1-r}).$$

Proof. Let $U_n = \max_{j \leq n} u_j$ and $m_n = \min_{j \leq n} u_j$. Nonnegativity gives

$$U_n \leq U_{n-1} \max(1, \sigma_n), \quad m_n \geq m_{n-1} \min(1, \sigma_n).$$

All row sums are positive. Beyond a finite index $|\sigma_n - 1| \leq 1/2$, and $\sum |\sigma_n - 1| < \infty$. Thus the upper product is finite and the lower product strictly positive. We have $0 < a \leq u_n \leq A_0 < \infty$ without assuming convergence.

For $n \geq 2$ set $d_n = u_n - u_{n-1}$ and, for $1 \leq j \leq n-2$, $W(n, j) = \sum_{k=j+1}^{n-1} w(n, k)$. Telescoping the individual term u_{n-k} gives the exact identity

$$d_n = (\sigma_n - 1)u_{n-1} - \sum_{j=1}^{n-2} W(n, j)d_{n-j}. \quad (28)$$

All difference indices are at least 2. Interchanging finite sums, its weighted norm is

$$T_r(n) = \sum_{j=1}^{n-2} W(n, j) \left(\frac{n}{n-j} \right)^r = \sum_{k=1}^{n-1} w(n, k) \sum_{j=1}^{k-1} \left(\frac{n}{n-j} \right)^r. \quad (29)$$

For $k \leq n/2$ the inner sum is at most $2^r k$, giving a summable majorant $\text{const } kR^k/k!$. For $k > n/2$ it is at most n^{r+1} , and the factorial tail makes that entire contribution tend to zero faster than any fixed inverse power. Fixed- k convergence therefore proves

$$T_r(n) \longrightarrow \sum_{k \geq 1} (k-1)p_k = \mu - 1 < 1. \quad (30)$$

Choose $\theta < 1$ and N with $T_r(n) \leq \theta$ for $n \geq N$. Boundedness of u bounds the forcing term in (28) by $B_0 n^{-r}$. Choose A_1 no smaller than $B_0/(1-\theta)$ and all finitely many initial $n^r |d_n|$. Induction in (28) gives

$$|d_n| \leq n^{-r} (B_0 + A_1 T_r(n)) \leq A_1 n^{-r}.$$

The differences are absolutely summable, so a limit C exists, and

$$|u_N - C| \leq A_1 \sum_{n > N} n^{-r} \leq \frac{A_1}{r-1} N^{1-r}. \quad (31)$$

The lower bound on u_n makes $C > 0$. □

For the Lengyel kernel, $\mu = 2L$ and the limiting norm is

$$2 \log 2 - 1 = 0.3862943611 \dots < 1.$$

The inequality, not the displayed decimal, is what the proof needs. Every weighted limit concerns a fixed r ; no uniformity for $r = r(n)$ is being used.

7 Completion and anchoring of the expansion

Proof of Theorem 1.1. For a fixed M , use the positively patched P_M and put

$$h_n = f_n P_M(n), \quad u_n = \frac{z_n}{h_n}, \quad w_M(n, k) = K(n, k) \frac{P_M(n-k)}{P_M(n)}.$$

These weights are nonnegative and inherit the factorial bounds up to constants. They converge to p_k for each fixed k . Dividing (24) by $P_M(n)$ gives $\sigma_n = 1 + O_M(n^{-M-2})$. Lemma 6.1 with $r = M + 2$ yields

$$\frac{z_n}{f_n P_M(n)} = C_M + O_M(n^{-M-1}), \quad C_M > 0.$$

Since $P_M(n) \rightarrow 1$, each C_M is the same limit of z_n/f_n . In particular they all equal the $M = 0$ constant C . Multiplication by the bounded P_M proves (4). The normalization $c_0 = 1$ and nonzero triangular multipliers prove uniqueness of all coefficients; the limiting normalization proves uniqueness of C . $\square$

This also explains why successive orders do not fit independent leading constants. The constant is the same already known Lengyel limit at every order.

The proof is effective in principle. If explicit bounds and a finite computation establish $|d_n| \leq A_M n^{-M-2}$ beyond an index, then

$$\left| C - \frac{z_N}{f_N P_M(N)} \right| \leq \frac{A_M}{M+1} N^{-M-1}. \quad (32)$$

The kernel remainder gives computable bounds, and finite-cutoff estimates make the weighted contraction effective. No numerically efficient A_M or evaluated interval certificate is claimed here. In particular a stable decimal extrapolation is not a substitute for (32) with verified inputs.

8 One complex marking disk for all fixed orders

We now prove (6). Positivity no longer holds for complex q ; a separate boundedness bootstrap is necessary.

Theorem 8.1 (Complex-uniform expansion). *On a neighborhood of the closed disk $D_0 = \{q : |q - 1| \leq 1/16\}$ define*

$$L(q) = \text{Log}(q+1) - \text{Log } q, \quad b(q) = 1 + L(q)/3, \quad f_n(q) = (n!)^2 [2L(q)]^{-n} n^{-b(q)},$$

using analytic logarithms in the right half-plane. There is an analytic limit $C(q) = \lim_n Z_n(q)/f_n(q)$ in the interior of D_0 , with uniform convergence on D_0 . On one smaller closed disk D about 1, C is nonzero and, for every fixed $M \geq 0$,

$$\frac{Z_n(q)}{C(q)f_n(q)} = \sum_{j=0}^M c_j(L(q))n^{-j} + O_{D,M}(n^{-M-1}) \quad (33)$$

uniformly. The same disk D works for every fixed M . Constants, finite patches, and onset indices may depend on M . On every smaller disk, each fixed derivative of the analytic remainder in (33) has the same order.

Proof. First, the explicit disk has an absolute contraction margin. Since $L'(q) = -1/[q(q+1)]$, integration along a radial segment gives

$$|L(q) - \log 2| \leq \frac{16}{465} \quad (q \in D_0).$$

The elementary inequality $\log 2 < 7/10$ implies $|L(q)| < 3/4$. (Already the first five terms of the exponential series show $e^{7/10} > 2$.) The limiting weights are

$$p_k(q) = q \frac{L(q)^k}{k!}, \quad \sum_{k \geq 1} p_k(q) = 1, \quad \sum_{k \geq 1} k p_k(q) = (q+1)L(q). \quad (34)$$

For $g(x) = (x-1)e^x + 1$, $g'(x) = xe^x \geq 0$, and $e^{3/4} > 2$. Therefore

$$\sum_{k \geq 1} (k-1)|p_k(q)| = |q|g(|L(q)|) < \frac{17}{16}g(3/4) < \frac{17}{32} < 1. \quad (35)$$

This disk selection contains no reference to M .

The kernel polynomials of Lemma 4.1 depend only on L and not separately on q . For $r \geq 1$ their moments now use

$$\sum_{k \geq 1} p_k(q) k^r = (q+1)T_r(L(q)), \quad (36)$$

whereas the zeroth moment remains 1. The q_1 moment is identically zero. Every nontrivial residual polynomial vanishes at $k=0$; its moment therefore has a factor $(q+1)L(q)$. The triangular multiplier is $m(q+1)L(q)$. These common factors cancel, leaving exactly the same $c_m \in \mathbb{Q}[L]$ from Section 5. This proves universality at every order, not just for the displayed coefficients.

For fixed M , the polynomial $P_M(n, q) = \sum_{j=0}^M c_j(L(q))n^{-j}$ tends uniformly to 1 on D_0 . Choose an M -dependent N_M beyond which $|P_M - 1| \leq 1/2$, and set $P_M = 1$ at the finitely many earlier integer indices. The patch is analytic in q and bounded above and away from zero. Put

$$h_n(q) = f_n(q)P_M(n, q), \quad u_n(q) = Z_n(q)/h_n(q), \quad w(n, k; q) = qS(n, n-k) \frac{h_{n-k}(q)}{h_n(q)}.$$

Lemma 3.1 and uniform bounds for the patch give constants A, B, R such that

$$|w(n, k; q)| \leq A \frac{R^k}{k!} \quad (k \leq n/2), \quad |w(n, k; q)| \leq An^B \frac{R^k}{k!} \quad (k < n), \quad (37)$$

uniformly on D_0 . One may take $R = \sup_{D_0} |L(q)|$ and absorb $|q|$ into A . Fixed- k convergence to $p_k(q)$ is uniform. The uniform Cauchy proof in Lemma 4.1, the moment cancellations, and the factorial tails give

$$\sigma_n(q) := \sum_k w(n, k; q) = 1 + O_{M, D_0}(n^{-r}), \quad r = M + 2. \quad (38)$$

The finite patch changes only factorially small tails.

Set $W(n, j; q) = \sum_{k > j} w(n, k; q)$. The triangle inequality, followed by exactly the near/far split in the proof of (30), yields

$$\limsup_{n \rightarrow \infty} \sup_{q \in D_0} \sum_{j=1}^{n-2} |W(n, j; q)| \left(\frac{n}{n-j} \right)^r \leq \sup_{q \in D_0} \sum_{k \geq 1} (k-1)|p_k(q)| < 1. \quad (39)$$

A finite- k cutoff makes the uniform convergence explicit; the remaining terms are dominated by a constant multiple of $kR^k/k!$, with the long-jump portion factorially negligible. Thus for this fixed r there are N and $\theta < 1$ bounding every weighted norm by θ beyond N .

It remains to prove boundedness without positivity. Identity (28) remains valid for the complex functions. Let B_0 bound $n^r|\sigma_n - 1|$ uniformly beyond N , and define

$$U_n = \max_{1 \leq j \leq n} \sup_{D_0} |u_j|, \quad D_n = \max_{2 \leq j \leq n} j^r \sup_{D_0} |u_j - u_{j-1}|.$$

If the maximum defining D_n occurs after the initial segment, (28) implies $D_n \leq B_0 U_n + \theta D_n$. If it occurs earlier it is D_{N-1} . Hence in all cases

$$D_n \leq D_{N-1} + \frac{B_0}{1 - \theta} U_n. \quad (40)$$

Independently, telescoping gives, with $S_N = \sum_{j \geq N} j^{-r}$,

$$U_n \leq U_{N-1} + S_N D_n. \quad (41)$$

Increase N while retaining the already obtained B_0, θ until $B_0 S_N / (1 - \theta) \leq 1/2$. Combining the two inequalities proves

$$U_n \leq 2(U_{N-1} + S_N D_{N-1})$$

for every n , and then also bounds D_n . The initial maxima are finite because they involve finitely many analytic functions on a compact disk. No boundedness of u was assumed to obtain this conclusion.

We have proved uniformly that $|u_n - u_{n-1}| \leq A_M n^{-M-2}$, so

$$u_n(q) = C_M(q) + O_{M,D_0}(n^{-M-1}).$$

Uniform convergence makes C_M holomorphic in the disk interior. Since $P_M \rightarrow 1$ uniformly, all C_M equal the same $M = 0$ limit C . Theorem 1.1 gives $C(1) > 0$. Continuity supplies a once-only smaller disk D on which C has no zeros. Dividing by this bounded-away-from-zero function proves (33). Cauchy's estimates on nested disks give the derivative assertion. $\square$

The factor $n^{-b(q)}$ always means $\exp(-b(q) \log n)$ with the real logarithm of the positive integer n . The factor $[2L(q)]^{-n}$ has an integer exponent, so it creates no additional branch choice. The order-independent disk does not mean order-independent estimates: the eventual contraction index and the patch can grow with M .

9 Chain length and all fixed cumulants

Let $\ell_j(L)$ be the formal logarithmic coefficients determined by

$$\log \left(\sum_{j \geq 0} c_j(L) x^j \right) = \sum_{j \geq 1} \ell_j(L) x^j. \quad (42)$$

Only finitely many coefficients are used at any one order. In particular,

$$\ell_1(L) = \frac{L}{6}, \quad \ell_2(L) = -\frac{L(8L^2 + 45L - 45)}{1620}.$$

On a sufficiently small fixed complex t -disk define

$$A(t) = \log \frac{L}{L(e^t)}, \quad B(t) = \frac{L - L(e^t)}{3}, \quad D(t) = \log \frac{C(e^t)}{C(1)}, \quad (43)$$

with logarithms zero at $t = 0$. Theorem 8.1 makes the normalized bracket uniformly close to 1 for sufficiently large n ; its analytic logarithm therefore has the usual fixed-order Taylor remainder. If a common zero-free disk for every finite n is desired, shrink once more for the finitely many remaining polynomials, which are nonzero at $q = 1$.

Corollary 9.1 (Logarithmic generating function). *For every fixed $M \geq 0$, uniformly on a fixed small complex disk,*

$$\begin{aligned} \log \mathbb{E} e^{tH_n} &= nA(t) + (\log n)B(t) + D(t) \\ &\quad + \sum_{j=1}^M (\ell_j(L(e^t)) - \ell_j(L))n^{-j} + O_M(n^{-M-1}). \end{aligned} \quad (44)$$

The remainder is analytic; every fixed derivative has the same order on smaller disks. Thus the r th cumulant, for each fixed $r \geq 1$, is

$$\kappa_r(H_n) = nA^{(r)}(0) + (\log n)B^{(r)}(0) + D^{(r)}(0) + \sum_{j=1}^M \left. \frac{d^r}{dt^r} \ell_j(L(e^t)) \right|_{t=0} n^{-j} + O_{M,r}(n^{-M-1}). \quad (45)$$

Proof. Divide the marked expansion by its value at $q = 1$ and take the analytic logarithm. The factorials cancel, the exponential scale gives nA , and the power of n gives $(\log n)B$. Finite logarithmic Taylor expansion supplies the ℓ_j terms with a uniform analytic remainder. Apply Cauchy's formula on a nested disk to differentiate that remainder. $\square$

Direct differentiation gives

$$\begin{aligned} A'(0) &= \frac{1}{2L}, \quad A''(0) = \frac{1-L}{4L^2}, \quad B'(0) = \frac{1}{6}, \quad B''(0) = -\frac{1}{12}, \\ \left. \frac{d}{dt} \ell_1(L(e^t)) \right|_0 &= -\frac{1}{12}, \quad \left. \frac{d^2}{dt^2} \ell_1(L(e^t)) \right|_0 = \frac{1}{24}. \end{aligned}$$

Taking $K_1 = D'(0)$ and $K_2 = D''(0)$ proves (7)–(8). These constants depend on the strict-transition convention $H_1 = 0$. Counting partitions in the chain instead would add 1 to the mean and its constant.

Corollary 9.2 (The known central limit theorem). *Put $v = (1 - L)/(4L^2) > 0$. Then*

$$\frac{H_n - \mathbb{E}H_n}{\sqrt{nv}} \implies \mathcal{N}(0, 1).$$

The denominator may also be replaced by the true standard deviation.

Proof. In (44) take $t = is/\sqrt{nv}$ for fixed real s and subtract the exact mean term. The quadratic contribution of nA tends to $-s^2/2$; its cubic and higher terms are $O(n|t|^3) = O(n^{-1/2})$. After centering, the analytic terms multiplied by $\log n$ or bounded coefficients contribute $o(1)$. Characteristic functions converge to $e^{-s^2/2}$, proving the limit. Equation (8) gives $\text{Var } H_n/(nv) \rightarrow 1$. $\square$

This deduction credits the earlier theorem of Van Cutsem–Ycart [5]. It is not accompanied here by a distributional error estimate or a local limit assertion.

10 Logarithmic asymptotics and inversion

Define

$$\beta = L/3, \quad K = 2\pi C, \quad \kappa = 1 + \frac{1}{2} \log(2L), \quad H(x) = 2x \log(x/e^\kappa).$$

Proposition 10.1 (The logarithmic coefficients). *For every fixed $M \geq 0$,*

$$\log z_n = H(n) - \beta \log n + \log K + \sum_{j=1}^M d_j n^{-j} + O_M(n^{-M-1}). \quad (46)$$

The d_j are obtained by adding the formal series in (42) to the odd Stirling series:

$$\sum_{j \geq 1} d_j x^j = \sum_{j \geq 1} \ell_j(L) x^j + \sum_{r \geq 1} \frac{B_{2r}}{r(2r-1)} x^{2r-1}. \quad (47)$$

Here the Bernoulli numbers use $B_2 = 1/6$. In particular,

$$d_1 = \frac{1+L}{6}, \quad d_2 = -\frac{L(16L^2 + 90L - 90)}{3240}.$$

Proof. At each fixed order, take the logarithm of the positive normalized expansion (4); Taylor's remainder is legitimate because the bracket tends to 1. Add the fixed-order Stirling expansion

$$2 \log(n!) = 2n \log n - 2n + \log n + \log(2\pi) + \sum_{r \geq 1} \frac{B_{2r}}{r(2r-1)} n^{-(2r-1)},$$

where the last series is truncated as needed with its fixed-order remainder. The $+\log n$ cancels the $-\log n$ in $-\beta \log n$, leaving exactly $-\beta \log n$. The remaining terms give (46)–(47). $\square$

10.1 A Lambert core and a triangular inverse

For $y \rightarrow +\infty$ let

$$w = W_0\left(\frac{y}{2e^\kappa}\right), \quad X = \frac{y}{2w}, \quad U = \log X, \quad \Lambda = 2(1+w) = 2U - \log(2L). \quad (48)$$

The principal real branch is used. Since $we^w = y/(2e^\kappa)$, we have $X = e^{\kappa+w}$, $H(X) = y$, and $H'(X) = \Lambda > 0$.

For every fixed M consider the smooth model

$$F_M(x) = H(x) - \beta \log x + \log K + \sum_{j=1}^M d_j x^{-j}. \quad (49)$$

It is strictly increasing for all sufficiently large x , since $F'_M(x) \sim 2 \log x$. Let $\xi_M(y)$ be its unique sufficiently large solution of $F_M(x) = y$.

Theorem 10.2 (Constructive fixed-order inverse). *Set $v_0(U) = (\beta U - \log K)/\Lambda$. The coefficients $v_m(U)$ for $m \geq 1$ are uniquely determined by requiring the following formal expression, with $t = 1/X$ and $\delta = \sum_{m \geq 0} v_m(U) t^m$, to vanish:*

$$\begin{aligned} 0 = & \Lambda \delta - \beta U + \log K + \sum_{r \geq 2} \frac{2(-1)^r}{r(r-1)} \delta^r t^{r-1} \\ & - \beta \log(1+t\delta) + \sum_{j \geq 1} d_j t^j (1+t\delta)^{-j}. \end{aligned} \quad (50)$$

At order $m \geq 1$, take minus the coefficient of t^m in all terms except $\Lambda\delta$, using $v_0, \dots, v_{m-1}$, and divide by Λ . Then

$$\xi_M(y) = X + \sum_{m=0}^M v_m (\log X) X^{-m} + O_M\left(\frac{X^{-M-1}}{\log X}\right). \quad (51)$$

Each v_m is a rational function of U over $\mathbb{Q}[L, \log K, \log(2L)]$; $v_0 = O(1)$ and $v_m = O(1/U)$ for $m \geq 1$.

Proof. Taylor expansion at X gives

$$\frac{H^{(r)}(X)}{r!} = \frac{2(-1)^r}{r(r-1)X^{r-1}} \quad (r \geq 2),$$

which produces the first nonlinear sum in (50). The other terms follow from the Taylor expansions of $\log(X + \delta)$ and $(X + \delta)^{-j}$. Every nonlinear term carries at least one factor of t ; hence v_m first appears as Λv_m at degree m . This proves triangularity, uniqueness, and the rational-function claim. Since $\Lambda \asymp U$, v_0 is bounded. Inductively the remaining degree- m expression is bounded for large U , proving $v_m = O(1/U)$.

For fixed M , substitute the finite displacement in F_M . The recursion cancels the residual through X^{-M} , and the remaining Taylor remainder is $O_M(X^{-M-1})$. All finitely many displacement coefficients are bounded; for large X the Taylor expansions are uniform on $|\delta|/X < 1/2$. On a fixed-size neighborhood of X , $F'_M(x) \sim 2 \log X$. The intermediate value theorem supplies a root there, eventual monotonicity makes it the unique large root, and the mean value theorem divides the residual by this derivative, proving (51). $\square$

The first terms are

$$\begin{aligned} v_1 &= -\frac{v_0^2 - \beta v_0 + d_1}{\Lambda}, \\ v_2 &= -\frac{(2v_0 - \beta)v_1 - v_0^3/3 + \beta v_0^2/2 - d_1 v_0 + d_2}{\Lambda}. \end{aligned} \quad (52)$$

In particular

$$\xi_1(y) = X + v_0 - \frac{v_0^2 - \beta v_0 + d_1}{X\Lambda} + O\left(\frac{1}{X^2 \log X}\right).$$

Although $v_0 \rightarrow \beta/2 = L/6$, replacing the full U -dependent v_0 by that limit discards a generally much larger $1/\log X$ correction and is not valid at the displayed precision.

10.2 Sampled values and the threshold staircase

The sequence is strictly increasing from $n = 2$ onward, because

$$z_{n+1} \geq S(n+1, n)z_n = \binom{n+1}{2} z_n > z_n \quad (n \geq 2). \quad (53)$$

The equality $z_1 = z_2 = 1$ is why range inversion is stated for $n \geq 2$.

Corollary 10.3 (Eventual recovery on sampled inputs). *If the input is known to be $y = \log z_n$, then*

$$\xi_M(y) = n + O_M\left(\frac{n^{-M-1}}{\log n}\right).$$

The finite expression in (51) has the same accuracy. Nearest-integer rounding therefore recovers n for all sufficiently large sampled inputs.

Proof. Equation (46) gives $F_M(n) = y + O_M(n^{-M-1})$. Around n the derivative is comparable to $\log n$. Apply the same mean-value residual argument as above; also $X \sim n$. The error eventually falls below $1/2$. $\square$

This is an asymptotic recovery theorem, not an implemented certified cutoff using approximate C .

For an arbitrary real threshold define

$$N(y) = \min\{n \geq 2 : \log z_n \geq y\}.$$

Theorem 10.4 (Two-ceiling threshold enclosure). *For every fixed M there are $A_M > 0$ and Y_M such that for $y \geq Y_M$, writing $\xi = \xi_M(y)$ and*

$$\varepsilon_M(y) = A_M \frac{\xi^{-M-1}}{\log \xi},$$

one has

$$\lceil \xi - \varepsilon_M(y) \rceil \leq N(y) \leq \lceil \xi + \varepsilon_M(y) \rceil. \quad (54)$$

The center can be replaced by the finite expression in (51) after increasing A_M . Choose Y_M also so that $2\varepsilon_M(y) < 1$. Then the two ceilings either agree or differ by one, giving at most two adjacent candidates.

Proof. Choose B_M, N_0 such that $|\log z_n - F_M(n)| \leq B_M n^{-M-1}$ for $n \geq N_0$. On $[\xi/2, 2\xi]$, for large ξ we have $F'_M(x) \geq c \log \xi$. Choose A_M large enough that this derivative times ε_M dominates $B_M n^{-M-1}$ throughout the interval. For every integer $n \geq \xi + \varepsilon_M$ there, $\log z_n \geq y$; for every integer $n < \xi - \varepsilon_M$ there, $\log z_n < y$. Integer nodes near $\xi/2$ and 2ξ , together with monotonicity, extend these statements outside the interval once the finite initial segment is excluded by enlarging Y_M . Taking ceilings gives (54). The inverse approximation error is of the same order as ε_M and is absorbed by enlarging its constant. Finally $\varepsilon_M \rightarrow 0$, so Y_M can be enlarged until the interval width is less than 1. $\square$

The strict lower comparison and non-strict upper comparison match the definition with $\geq y$. Even an arbitrarily accurate asymptotic approximation need not have the correct ceiling at every real threshold near a jump. Thus no single-ceiling formula without an error envelope is asserted. The constants A_M, Y_M have not been numerically evaluated or certified; the exact integer inverse in the code avoids them entirely within its declared resource bound.

Remark 10.5 ([write] Section 10 in the terms of the transseries volume, 5 October 2026). The source cites the transseries collection [10] for “the Lambert core, residual transport, and staircase qualifications”. Statement by statement, against the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`:

(a) *The core.* The core (48) is an instance of `p0:prop:factorial-core`, not of `p0:thm:lambert-core`. The equation $H(X) = y$ reads $X(\kappa' \log X + d') = y$ with $\kappa' = 2$ and $d' = -2\kappa$. The proposition gives $v = W_0((y/\kappa')e^{d'/\kappa'}) = W_0(y/(2e^\kappa)) = w$, $X = \exp(v - d'/\kappa') = e^{\kappa+w}$ and core slope $\kappa'(v+1) = 2(1+w) = \Lambda$, exactly as in (48); its side condition $\kappa' \log X + d' > 0$ is $w > 0$, which holds for $y > 0$. The volume’s Lambert core proper concerns the linear–logarithmic phase $aX + b \log X$; the phase $2X \log X$ here is outside its exponential–power model, as the volume says of the double factorial.

(b) *The recursion.* The recursion (50) is an instance of `p0:thm:core-reversion` after the change of variable $E = t\delta = \delta/X$ (the displacement measured in units of X). Multiplying (50) by t and using $\sum_{r \geq 2} (-1)^r u^r / (r(r-1)) = (1+u) \log(1+u) - u$ gives

$$\Lambda E + h(E) + f(t, E) = 0, \quad h(u) = 2((1+u) \log(1+u) - u),$$

$$f(t, u) = t \left(\log K - \beta U - \beta \log(1+u) + \sum_{j \geq 1} d_j t^j (1+u)^{-j} \right),$$

with $h \in u^2 R[[u]]$ and $f \in tR[[t, u]]$ over the commutative $\mathbb{Q}$ -algebra $R = \mathbb{Q}[L, \log K, \log(2L), U][\Lambda^{-1}]$ of real rational functions of U , in which $\Lambda = 2U - \log(2L)$ is invertible. The theorem's unique solution $E = \sum_{n \geq 1} e_n t^n$ has $e_{m+1} = v_m$, and its triangular recurrence is the source's rule “take minus the coefficient of t^m ... and divide by Λ ”; for instance $e_1 = -\Lambda^{-1}[t^1]f = (\beta U - \log K)/\Lambda = v_0$. This is the same shape as the volume's double-factorial instance (`p0:rem:core-instances`), with the factor 2 in h and a different f .

(c) *The analytic step.* The analytic step of Theorem 10.2 (a root of $F_M = y$ near the finite expression, then division of the residual by the slope) is the argument of `p0:thm:backward-error`, applied to F_M on $[X-1, X+1]$, where $F'_M \asymp \log X$; the source proves it directly.

(d) *Sampled recovery.* The rounding in Corollary 10.3 is `p0:thm:staircase` (3) for $A_n = \log z_n$ with $n_1 = 2$ (strictly increasing and unbounded by (53)), for every admissible interpolation, since $\nu(\log z_n) = n$; the input bound $|\xi_M(\log z_n) - n| < \frac{1}{2}$ is the source's own estimate.

(e) *The threshold enclosure.* Theorem 10.4 is proved directly, by comparing $\log z_n$ with F_M at the integers; as proved it is not an instance of `p0:thm:staircase`, which is stated for an interpolated inverse. Its conclusion also follows from `p0:thm:staircase` (1) with one interpolation chosen for the purpose (a second route, with possibly different constants). *Proof.* Put $r_n = \log z_n - F_M(n)$. By (46) there are B_M and N_0 with $|r_n| \leq B_M n^{-M-1}$ for $n \geq N_0$. Since $F'_M(x) \sim 2 \log x$, choose an integer $n_1 \geq \max(N_0, 3)$ with $F'_M(x) \geq \log x$ for $x \geq n_1$ and $2^{M+1} B_M n_1^{-M-1} < \log n_1$. On $[n_1, \infty)$ let ρ be the piecewise-linear function with $\rho(n) = r_n$ at the integers, and $\mathcal{A} = F_M + \rho$. Then $\mathcal{A}$ is continuous, $\mathcal{A}(n) = \log z_n$, and on each $[n, n+1]$

$$\mathcal{A}' \geq \log n - |r_{n+1} - r_n| \geq \log n - 2B_M n^{-M-1} > 0,$$

because $2B_M n^{-M-1} \leq 2^{M+1} B_M n^{-M-1}$, which decreases in n , while $\log n$ increases. So $\mathcal{A}$ is strictly increasing and unbounded, an admissible interpolation of `p0:def:three-inverses`, and for $x \in [n, n+1]$, $|\mathcal{A}(x) - F_M(x)| \leq B_M n^{-M-1} \leq 2^{M+1} B_M x^{-M-1}$. For large y put $\nu = \mathcal{A}^{-1}(y)$ and $\xi = \xi_M(y)$, both at least n_1 . Since $F_M(\xi) = y = \mathcal{A}(\nu)$, the mean value theorem gives

$$|\nu - \xi| \log \min(\nu, \xi) \leq |F_M(\nu) - F_M(\xi)| = |F_M(\nu) - \mathcal{A}(\nu)| \leq 2^{M+1} B_M \nu^{-M-1}.$$

The right side is at most $2^{M+1} B_M n_1^{-M-1} < \log n_1 \leq \log \min(\nu, \xi)$, so $|\nu - \xi| < 1$; for $\xi \geq 3$ this gives $\nu^{-M-1} \leq (\xi - 1)^{-M-1} \leq 2^{M+1} \xi^{-M-1}$ and $\log \min(\nu, \xi) \geq \log(\xi - 1) \geq \frac{1}{2} \log \xi$, hence $|\nu - \xi| \leq \varepsilon := 2^{2M+3} B_M \xi^{-M-1} / \log \xi$. For $y \geq \log z_{n_1}$, `p0:thm:staircase` (1) gives $\min\{n \geq n_1 : \log z_n \geq y\} = \lceil \nu \rceil$, and this minimum is $N(y)$, because $\log z_n < \log z_{n_1} \leq y$ for $2 \leq n < n_1$ by (53). As the ceiling is nondecreasing, $\lceil \xi - \varepsilon \rceil \leq N(y) \leq \lceil \xi + \varepsilon \rceil$, which is (54) with $A_M = 2^{2M+3} B_M$ for all sufficiently large y . $\square$

The ordinary piecewise-linear interpolation of $\log z_n$ would not do here: between the integers it differs from F_M by about $F''_M/8 \approx 1/(4x)$, which gives $|\nu - \xi|$ only of order $1/(x \log x)$. When $2\varepsilon < 1$, `p0:prop:certified-rounding` (one residue class) says that at most two exact evaluations of z_n decide $N(y)$, which is the source's “at most two adjacent candidates”. No novelty is claimed for any of the inversion.

11 Exact computation and reproducibility

The accompanying package has three deliberately separate layers.

1. The standard-library core computes exact integers, chain polynomials, fixed moments and cumulants, and a resource-bounded exact integer threshold inverse. It does not require a numerical value of C .
2. An optional SymPy program constructs kernel and correction coefficients by the finite recipe in Section 5, checks logarithmic and cumulant algebra, and checks formal inverse cancellation. Its output is exact symbolic algebra.
3. An optional mpmath program evaluates normalized sequences and centered moments at finite n . Its decimal outputs are diagnostics, not interval enclosures or proofs of error bounds.

All public checks use explicit exceptions rather than assertions, so optimized Python execution does not remove the checks. Input lengths and resource parameters are validated before expensive operations. Computed large integers are converted to decimal in bounded chunks without changing Python’s process-wide integer conversion limit. The package tests genuinely large output through the real command-line interface with `PYTHONINTMAXSTRDIGITS=640`. No global recursion, integer-conversion, or precision settings are changed; optional numerical work uses a local mpmath context.

11.1 Independent finite checks

Full partition-poset enumeration is feasible at small n and provides a check independent of the rank-only recurrence. Enumerate all set partitions of $[n]$, test refinement for every pair, and sum chains over all strictly coarser partitions. This gives

$$\begin{aligned}
 Z_1(q) &= 1, \\
 Z_2(q) &= q, \\
 Z_3(q) &= q + 3q^2, \\
 Z_4(q) &= q + 13q^2 + 18q^3, \\
 Z_5(q) &= q + 50q^2 + 205q^3 + 180q^4, \\
 Z_6(q) &= q + 201q^2 + 1865q^3 + 4245q^4 + 2700q^5.
 \end{aligned}$$

The numbers of poset elements are 1, 2, 5, 15, 52, 203. These results also check the exact length convention.

For exact moment computations one may use ordinary derivatives $R_r(n) = Z_n^{(r)}(1)$:

$$R_r(n) = \sum_{j < n} S(n, j) (R_r(j) + r R_{r-1}(j)), \quad (55)$$

with $R_0(1) = 1$, higher derivatives zero, and the second summand absent when $r = 0$. Alternatively, Euler derivatives give raw moments directly by the binomial theorem applied to the additional first step. Stirling conversion relates these two implementations. Dividing the exact sums by z_n gives rational moments, and the standard triangular moment–cumulant identity gives exact rational cumulants. The package checks both against the chain polynomials and the full poset enumeration.

A further exact check is the inequality $2^k k! S(n, n-k) \leq ((n)_k)^2$ from Lemma 3.1. Exact inverse checks verify both attainment and minimality: if the bounded search returns $n > 2$, then $z_{n-1} < T \leq z_n$ for its positive integer input T .

11.2 Numerical diagnostics and their limits

For reference, numerical substitution of $L = \log 2$ in the exact coefficients gives

$$\begin{aligned} c_1 &\approx 0.1155245300933242, & c_2 &\approx 0.0109365636395843, \\ c_3 &\approx -0.0238254567258004, & c_4 &\approx -0.0275946778045995. \end{aligned}$$

Finite- n corrected normalizations are consistent with $C \approx 1.098685805525187$. The following centered quantities from exact integer moment sums illustrate the smaller terms:

$$M_n = \mathbb{E}H_n - \frac{n}{2L} - \frac{\log n}{6} + \frac{1}{12n}, \quad V_n = \text{Var } H_n - \frac{1-L}{4L^2}n + \frac{\log n}{12} - \frac{1}{24n}.$$

n	M_n	V_n
100	-0.56552859348282824923	-0.22033553105629492173
200	-0.56552929520692838409	-0.22033372829125359404
400	-0.56552946656943563565	-0.22033328277962673835
600	-0.56552949796082401647	-0.22033320071484707943

These are compatible with $K_1 \approx -0.565529523$ and $K_2 \approx -0.220333135$. Neither these estimates nor the displayed estimate for C is interval-certified. Their role is to detect algebraic or indexing mistakes; they are not used in any theorem proof or exact threshold decision.

The README gives the exact commands, supported bounds, dependency versions, source links, checksums, and deterministic rebuild procedure. The PDF has no creation date or trailer identifier; the archive uses a fixed member order, timestamp, permissions, and compression policy. A full rebuild is intended to preserve every archive member and the entire archive byte for byte under the recorded toolchain, in ordinary and optimized Python modes and with the supported integer conversion cap. Cross-version TeX byte identity is not promised. No third-party source PDFs or private research notes are included.

[write] The package here, and the intake reruns, 5 October 2026. “The README” and “the archive” above are the delivered ones. In this report the four programs are `code/lengyel_exact.py`, `code/test_lengyel_exact.py`, `code/symbolic.py` and `code/diagnostics.py` (delivered in `code/`); the rebuild tool `code/build.py` was delivered at the package root; the recorded outputs `generated/*` are `data/generated-*`; `SOURCES.json` and `requirements-optional.txt` are in `data/`. The delivered PDF and SHA256SUMS are not shipped, and the report README replaces the delivered one. `code/build.py` expects the delivered layout (`Report223.tex`, `SHA256SUMS`, `generated/`), so it does not run in this directory; the report README gives a route from the arrival commit’s archive. The four programs need no layout: the test program finds `lengyel_exact.py` beside itself. At placement, on copies, the three programs regenerated `generated/*.txt` byte for byte, up to the CRLF line ends of Windows standard output. At the write, on copies of the shipped `code/` (Windows, Python 3.14.4; SymPy 1.14.0 and mpmath 1.3.0 as pinned), the test program printed all eight PASS lines in 6 s, the symbolic program its coefficients and three PASS lines in 106 s, and the diagnostics program in 3 s; after CRLF to LF each output equals the shipped `data/generated-*.txt` byte for byte. The full rebuild (`build.py`), which compares pdfTeX builds and archive bytes under the recorded TeX Live toolchain, was not run.

12 Further questions within a clear boundary

The arguments leave several concrete directions open without assuming their answers.

1. **Certified constants.** Turn the constructive bounds behind (32) into practical interval enclosures for C , then extend them to derivatives of $\log C(e^t)$. This would support evaluated inverse enclosures rather than existence constants alone.

2. **Dependence on the order.** Quantify how C_e , the Taylor remainder constants, the finite patches, and the weighted contraction onset grow with M . Such estimates are prerequisites for a justified late-order or optimal-truncation analysis; the fixed-order theorem itself supplies none.
3. **A larger marking region.** The explicit disk $|q - 1| \leq 1/16$ is convenient rather than optimal. A larger analytic domain would require tracking the absolute contraction and possible zeros of $C(q)$. The local proof does not establish a global marked-parameter result.
4. **Distributional refinements.** A local limit theorem, a quantitative normal approximation, or large deviations would require additional arguments not supplied here. Local limits and global large-deviation statements may also require control outside the small complex neighborhood. The cumulant expansion alone should not be promoted to those conclusions.
5. **Comparison with analytic iteration.** Relate the anchored recurrence limit and its marked derivatives directly to the contour constants in the prior analytic-iteration treatment, and clarify the full contents of the unrecovered Flajolet–Salvy manuscript if it becomes available.
6. **Aggregate versus depth asymptotics.** Compare this direct aggregate proof with consequences obtainable by summing proportional-depth iterated-Bell asymptotics, with particular attention to tail control and uniformity in depth. Their relationship deserves an actual derivation rather than an absence claim.

These are research questions, not additional asserted theorems. The completed result of this report is the explicit fixed-order recurrence expansion, its local analytic marking, the cumulant consequences, and the specialized inverse bounds proved above.

12.1 Further questions and research

[write] **Added 5 October 2026, batch 106.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question, with its source, the argument offered and what is missing. The six items above are Questions 1–6 below, in the same order; Questions 7 and 8 collect two statements the source makes elsewhere. The intake found no false claim in the source; the one error it records is in the OEIS entry (Remark 2.1).

1. **Certified constants** (source, item 1; Section 7: “The proof is effective in principle” and “The kernel remainder gives computable bounds, and finite-cutoff estimates make the weighted contraction effective”). Proved: (32) converts a verified bound $|d_n| \leq A_M n^{-M-2}$ into an enclosure of C . Not supplied: any explicit A_M , onset or interval value of C , K_1 , K_2 or the marked derivatives; the effectivity sentences are a sketch. The decimals $C \approx 1.098685805525187$, $K_1 \approx -0.565529523$, $K_2 \approx -0.220333135$ are diagnostics (Section 11; the dossier’s recomputation agrees for C to about $5 \cdot 10^{-13}$). By Theorem 1.1, $z_n/f_n = C(1 + L/(6n) + O(n^{-2}))$, which answers the convergence-speed question at the end of Babai and Lengyel’s Section 2 [2] (as the source reports it; not read by the write) asymptotically, but not with explicit constants. Missing: explicit kernel and contraction constants, then interval arithmetic.
2. **Dependence on the order** (source, item 2; Section 1.1). Every statement is for fixed M ; convergence, Gevrey bounds, optimal truncation and exponentially small terms are not addressed. Missing: growth of C_e , the Cauchy constants, the patches and the contraction onset in M .

3. **A larger marking region** (source, item 3; Theorem 8.1). The disk $|q - 1| \leq 1/16$ comes from the explicit contraction bound (35); the further shrink avoids zeros of $C(q)$, whose location is unknown. Missing: the contraction and the zero set of $C(q)$ on larger domains.
4. **Distributional refinements** (source, item 4). A local limit theorem, a rate in the central limit theorem, Edgeworth terms or large deviations for H_n are not proved; the cumulant expansion of Corollary 9.1 is local in t . Missing: control of $Z_n(q)$ away from $q = 1$ (for a local limit theorem, on the whole unit circle).
5. **Comparison with analytic iteration, and priority** (source, item 5, and Section 2). As the source reports it, Prellberg's presentation [3] gives the leading equivalent with a contour expression for the constant; once the normalizations are matched, and provided that equivalent holds as stated, that expression and the recurrence limit (3) are the same number, $\lim z_n/f_n$ (the write did not read the presentation). What is open is a direct derivation of one from the other, the marked derivatives K_1, K_2 in contour form, and the contents of the unfinished 1990 Flajolet–Salvy manuscript (the OEIS attributes the decimal of C to Flajolet and Salvy). Until then no first occurrence of $c_2, c_3, \dots$ is claimed. Missing: the sources.
6. **Aggregate versus depth asymptotics** (source, item 6). The exact identity $z_n = \sum_{m \geq 0} 2^{-m-1} H(n, m)$ of `a139383-iterated-bell-diagonals` (Part II, Section 17.2), where $H(n, m) = n! [z^n] (e^z - 1)^{om}$ are the iterated Bell numbers, links this report to the proportional-depth theorem `ibd:pd:thm:main` there. *Heuristic* [write]: its leading form `ibd:pd:eq:factorial` reads $H(n, m) \approx (n-1)! 2^{-n} m^n m^{-\beta/3} I(\beta)$ with $\beta = n/m$ and that report's amplitude I . The weight $2^{-m} m^n$ peaks at $m = n/L$ ($\beta = L$, slope $\lambda = 1/L \approx 1.443$), with $\sum_m 2^{-m} m^n \approx (n/(eL))^n \sqrt{2\pi n}/L$ by Laplace's method, and $n^n e^{-n} \sqrt{2\pi n} \sim n!$; formally, then,

$$z_n \approx (n!)^2 (2L)^{-n} n^{-1-L/3} \cdot \frac{L^{L/3-1} I(L)}{2}, \quad \text{that is} \quad C \stackrel{?}{=} \frac{L^{L/3-1} I(L)}{2} = \frac{D(1/L)}{2\sqrt{2\pi} L},$$

with D as in `ibd:pd:eq:amplitude`. The scale f_n , including the exponent $b = 1 + L/3$, comes out exactly. *Observation, not a claim.* The write estimated $I(L)$ from exact values of $H(n, m)$ (by the recurrence) at the two integers m nearest n/L , dividing by $(n-1)! 2^{-n} m^n m^{-n/(3m)}$ and interpolating linearly in β to $\beta = L$. This gives $I(L) \approx 1.66277, 1.66098, 1.66009$ and $L^{L/3-1} I(L)/2 \approx 1.102044, 1.100861, 1.100269$ at $n = 200, 300, 400$; these exceed C by 3.36, 2.18, 1.58 thousandths, about $0.65/n$, and two-point extrapolation in $1/n$ gives 1.098494 (from $n = 200, 300$) and 1.098492 (from $n = 300, 400$), about $1.9 \cdot 10^{-4}$ below C . The gap is the logarithmic correction of that theorem: by `ibd:pd:eq:P1` with $\delta = 0$, $\beta = L$, the first correction P_1/n contains $-\frac{L^2}{18} (\log n)^2/n$, which extrapolation in $1/n$ does not remove. Fitting $A + B/n + D_1 \log n/n + D_2 (\log n)^2/n$ to the sixteen values $n = 100, 120, \dots, 400$ (verification run, 5 October 2026) returns $D_2 = -0.02937$, against the predicted $-L^2 C/18 = -0.02933$, and $A = 1.098694$, within 10^{-5} of C ; with D_2 fixed at its predicted value, A agrees with C to $3 \cdot 10^{-6}$ ($n \geq 200$). These are fits to numerics, not a proof. (Sentence replaced after the independent check of 5 October 2026; the first wording read “about $1.9 \cdot 10^{-4}$ below C , a gap attributable to the logarithmic corrections of that theorem”.) Missing for a proof: that theorem is uniform only for slopes in compact sets and rests on the contour foundation of that report's Part I, so the sum needs separate bounds for $m \leq \varepsilon n$ and $m \geq n/\varepsilon$ (note $H(n, m) \leq H(n, m+1)$, since $S(n, n) = 1$); its corrections $P_j(\log n; \lambda, \delta)$ are polynomials in $\log n$ whereas (4) has none, which a summation would have to explain (at the level of $(\log n)^2/n$ it does: the tilt $m^{-n/(3m)}$ of the Laplace window contributes a factor $\exp(\frac{L^2}{18} (\log n)^2/n + \dots)$,

which cancels the term of P_1 above; the $\log n/n$ terms were not compared; added after the independent check of 5 October 2026); and $I(L)$ is not certified (that report certifies only $\beta = 1$). If proved, the identity would express C by that report’s contour functional, which bears on Question 5. Conversely, nothing here gives a pointwise depth equivalent from the aggregate, as that report says.

7. **An evaluated inverse** (Section 10, after Theorem 10.4: “The constants A_M, Y_M have not been numerically evaluated or certified”). Theorem 10.4 and Corollary 10.3 are existence statements; the shipped exact inverse is a bounded search. Missing: explicit B_M in (46) and a certified C (Question 1).
8. **Van Cutsem and Ycart’s variance conjecture** (Section 2). Their general conjecture for a variance equivalent, under hypotheses broader than this model ([5], p. 997, as the source reports it), is not resolved; Corollary 9.1 proves the variance expansion (8) for this model only. Missing: an argument for their general class.

[write] **Independent check of the write, 5 October 2026.** An adversarial check made by the intake after the write (commit 44ebdfef9) found every item valid and no mathematical error. It confirmed the live entry A005121 (revision #79) and its formula line verbatim, and, paging all 79 revisions of the history, that the line first appears in revision #8 (13 September 2003) as “(Lengyel)”, that revision #30 (18 April 2014) changed only the attribution to “[Lengyel]”, and that the sign is unchanged since 2003. Chains counted directly from the definition (refinement chains from the finest to the coarsest set partition) give $z_1, \dots, z_6 = 1, 1, 4, 32, 436, 9012$; the recurrence reproduces these and all 19 live data terms; and the printed line, solved with generic coefficients, has the unique solution $-\mathcal{Z}$ through z^7 . Applied to $\mathcal{Z}$ itself the printed right-hand side equals $\mathcal{Z} - z$ exactly, so the line fails only at the coefficient of z (wording of Remark 2.1 sharpened, with a dated note). It confirmed the five parts of Remark 10.5 against the transseries volume, including a line-by-line check of the second route to Theorem 10.4 in part (e). In Question 6 every printed digit reproduces, and the $1.9 \cdot 10^{-4}$ gap is explained by the term $-\frac{L^2}{18}(\log n)^2/n$ of **ibd:pd:eq:P1**: a free fit recovers $D_2 = -0.02937$ against the predicted -0.02933 , and the constant then agrees with C to about 10^{-5} (sentence replaced, with a dated note). From exact z_n to $n = 400$, an eight-node fit $z_n/f_n = C(1 + \sum_{k \leq 7} a_k n^{-k})$ on $n = 225, \dots, 400$ gives $C = 1.09868580552518701$, and $a_1, \dots, a_4$ agree with $c_1, \dots, c_4$ of (5) to 12, 12, 9 and 7 digits.

References

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- [8] OEIS Foundation, *A005121*, Lengyel numbers. Sequence entry; official raw source. The inspected source header is revision #79, 30 May 2026, 16:39:47; its formal generating-function sign is discussed in Section 2.
- [9] V. Reshetnikov, ProveIt, *Iterated Bell diagonals*, article in the OEIS sequence asymptotics collection, inspected at commit 9f19e58de1fa83abdc8504c166975d724d8a9791. Pinned article source. The compared blob is ab5014ecd6366ef621d4d45393609bf177c88339.
 [write, 5 October 2026: that report has no author line (its title block reads “A merged research report built from two manuscripts”). It is a139383-iterated-bell-diagonals in this collection, titled “Asymptotics of iterated Bell diagonals (OEIS A139383, A261280)”]; its article blob at the commit of this write is still ab5014ec.]
- [10] V. Reshetnikov, ProveIt, *Transseries and Inversion* and companion combinatorial material, repository revision 9f19e58de1fa83abdc8504c166975d724d8a9791. Pinned transseries collection. The Lambert core, residual transport, and staircase qualifications are used here as established general methods.